BEYOND THE JOKE: UNMASKING HATEFUL MEMES WITH CARA (/kɑːrə/)

(Contrastive **A**dversarial **R**easoning **A**rchitecture**)**

CS4248 - 30

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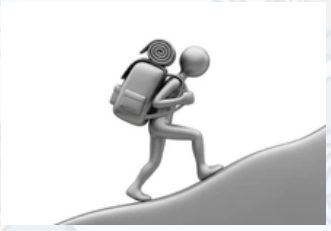
WHY?

Hateful memes **fuel real-world harm**, with millions shared daily targeting vulnerable individuals and communities

SotA models achieve only ~65% accuracy on hateful meme benchmarks, struggling with benign confounders and **context-dependent hate**

SEMANTIC FLUIDITY



"Heavy Lifting"

Can mean *"Carry the team to success"* or *"Bearing and unfair mental load"*

Note: We exploit Memecap's metaphors and meanings structure as it allows fluid mapping of either benign or adversarial meanings

EXAMPLES

THE CARA MODEL

I → **LMM** Qwen3 8B Thinking

$H_b = f_{bLMM}(I)$

$H_h = f_{hLMM}(I)$

Benign Interpretation

Metaphor: Child with pink bow in hair

Meaning: Casual depiction of a child's hairstyle, reflecting everyday fashion choices without gendered implication

Hateful Interpretation

Metaphor: Pink bow in child's hair

Meaning: Reinforces the idea that children's hairstyles should conform to traditional gendered beauty norms, potentially normalizing early gendered presentation

H_b, H_h → **RoBERTa Judge**

OCR Text: *"bows here, bows there, bows everywhere"* → T

$pred = f_{judge}(H_b, H_h, T)$

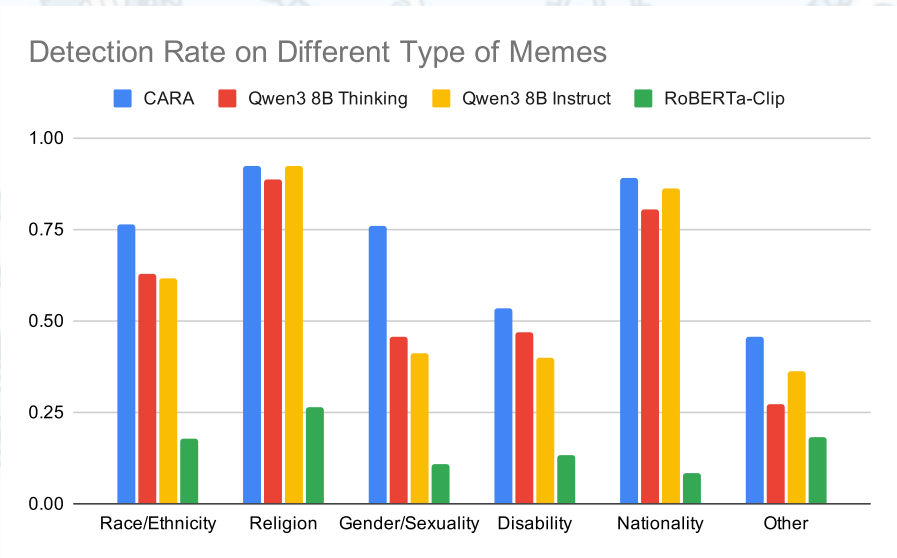
Label: Benign

CARA generates **contrastive benign and adversarial interpretations** of a meme using an LMM, then **combines them with OCR text** in a fine-tuned RoBERTa model to **classify** hateful intent, reducing the problem from multimodal fusion to textual entailment.

RESULTS AND ANALYSIS

Model	Accuracy ↑	AUROC ↑
CARA	**0.738**	**0.807**
Qwen3 8B Instruct only	0.664	0.664
Qwen3 8B Thinking only	0.658	0.658
RoBERTa-Clip only	0.520	0.595
Alternative Results (Ask us about it!)		
Qwen3 8B w/ SFT	0.728	0.728
Qwen3 8B w/ RAG	0.614	0.614

Note: The **AUROC** metric measures the ability to distinguish between classes across all thresholds!



CARA consistently performed the best in detecting hateful memes across different target groups

TRAINING METHODOLOGY

$D = (I_i, T_i, y_i)$

Facebook Dataset

I_{fb} *Full meme image* → **LMM** Qwen3 8B Thinking

$Hateful = H_h \quad Benign = H_b$

Contextual Reasoning: {reasoning}
Hateful Interpretation: {hateful_str}
Benign Interpretation: {benign_str}"

OCR Meme Caption T_{fb}

Target Label y

→ **RoBERTa** → p

Loss Function: ***Binary Cross Entropy Loss (BCE)***

$$L = -[y \log(p) + (1 - y) \log(1 - p)]$$

Rationalisation over Reasoning

Highly charged words can make LMMs use Chain of Thought not to actually explore the logic, but to generate steps to justify the conclusion it already wants to make

Finding truth through opposites

By training the judge model on adversarial pairs, we force the model to abandon lazy shortcuts and evaluate the true context of media

More context is not the cure

RAG fails on visual metaphors because injecting multiple definitions just creates noise. Instead of grounding the LMM, the extra context overwhelms the visual data and causes further confusion